

IE 221 Probability | Fall 2025–2026

Teamwork 4: SLLN & CLT Simulations and Monte Carlo π Estimation

Group Name: IE221-TeamWork-Group17

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1. Introduction

This project aims to demonstrate two fundamental results in probability theory—the **Strong Law of Large Numbers (SLLN)** and the **Central Limit Theorem (CLT)**—through Monte Carlo simulation. These theorems are essential tools for understanding the behavior of random variables when the sample size increases, and they constitute the theoretical foundation for many real-world applications in statistics, engineering, finance, and machine learning.

The **Strong Law of Large Numbers (SLLN)** explains why long-run averages stabilize. Specifically, it states that the sample mean converges almost surely to the expected value of a random variable. In practice, this theorem justifies why using larger datasets makes estimates of averages (e.g., average demand, mean lifetime, average cost) more reliable.

The **Central Limit Theorem (CLT)**, on the other hand, describes the distributional behavior of normalized sums (or sample means). It states that, under mild conditions, the distribution of standardized sums approaches the normal distribution as sample size increases. The CLT explains why normal approximations are widely applicable, and why the normal distribution appears frequently in natural and engineered systems.

In addition to SLLN and CLT simulations, this project includes a **Monte Carlo estimation of π** , which demonstrates how randomness and repeated sampling can approximate numerical constants. This part also reinforces the practical importance of probabilistic simulation techniques.

The scope of this study is:

- to simulate the SLLN using i.i.d. samples from **Uniform(0,1)**,

- to simulate the CLT using standardized sums and visualize convergence to $\mathbf{N(0,1)}$,
- to estimate π using Monte Carlo geometry,
- and to compare SLLN and CLT in terms of convergence type and experimental methodology.

2. Theoretical Background

2.1 Strong Law of Large Numbers (SLLN)

Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with finite expected value:

$$E[X_1] = \mu, E[|X_1|] < \infty$$

Define the sample mean as:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

The **Strong Law of Large Numbers** states:

$$\bar{X}_n \xrightarrow{a.s.} \mu \text{ as } n \rightarrow \infty$$

This means that for almost every sample path (with probability 1), the sequence of sample means approaches the true mean.

In our project, we generated i.i.d. samples from **Uniform(0,1)**. For $X \sim U(0,1)$:

$$\begin{aligned} \mu &= E[X] = 0.5 \\ \text{Var}(X) &= \sigma^2 = \frac{1}{12} \end{aligned}$$

The simulation computes cumulative means:

$$S_n = \frac{X_1 + X_2 + \dots + X_n}{n}$$

and investigates how quickly S_n becomes close to 0.5.

2.2 Central Limit Theorem (CLT)

Let $X_1, X_2, \dots$ be i.i.d. random variables with:

$$E[X_1] = \mu, \text{Var}(X_1) = \sigma^2 < \infty$$

Define the sum:

$$S_n = \sum_{i=1}^n X_i$$

The **Central Limit Theorem** states that the standardized sum converges in distribution to a standard normal random variable:

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{d} N(0,1) \text{ as } n \rightarrow \infty$$

Equivalently, the standardized sample mean can also be written as:

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$$

For $X \sim U(0,1)$, we again have:

$$\mu = 0.5, \sigma = \sqrt{1/12}$$

In our CLT simulation, for each chosen sample size n , we generate many independent replications, compute standardized sums Z_n , and examine whether their distribution resembles $N(0,1)$ using histograms and Q-Q plots.

2.3 Monte Carlo Method

The **Monte Carlo method** is a computational technique that approximates numerical values by repeated random sampling. It is particularly useful when:

- analytical solutions are difficult,
- integrals are high-dimensional,
- probability distributions are complex.

The core idea is that sample averages converge to expected values, which is justified by laws of large numbers.

In this project, Monte Carlo simulation is used in three contexts:

1. SLLN: cumulative sample mean converges to $E[X]$
2. CLT: distribution of standardized sums approaches normality
3. π estimation: geometric probability approximates π

3. Modes of Convergence

Although SLLN and CLT both describe limiting behavior as $n \rightarrow \infty$, they are fundamentally different theorems because they rely on different modes of convergence.

3.1 Theoretical Comparison

Theorem	Convergence Type	Meaning
SLLN	Almost sure convergence	Sample paths converge to μ with probability 1
CLT	Convergence in distribution	Distributions converge to $(N(0,1))$

Almost sure convergence is a strong type of convergence. It implies that for almost every possible realization of the random experiment, the computed sample mean will converge to the true mean.

Convergence in distribution is weaker. It does not claim that individual sample paths converge to a fixed constant. Instead, it states that the **shape of the distribution** of standardized sums approaches the standard normal distribution.

Therefore:

- **SLLN is a statement about a single sequence** $\bar{X}_1, \bar{X}_2, \dots$
- **CLT is a statement about distributions** of Z_n for increasing n

3.2 Experimental Comparison

The difference in convergence types directly affects how simulations must be performed.

SLLN Simulation Approach

For SLLN, it is sufficient to generate **one long sequence** of observations and compute the cumulative mean S_n . A line plot of S_n versus n is enough to observe convergence.

Reason: SLLN describes convergence **along a single sample path**. One realization already represents how averages behave in the long run.

CLT Simulation Approach

For CLT, one long sequence is not enough. The CLT describes the distribution of standardized sums, which requires:

- repeating the experiment **many times** for fixed n ,
- collecting many values of Z_n ,
- visualizing their distribution via histograms and Q-Q plots.

Reason: CLT is about **distributional behavior**. To approximate distribution empirically, a large number of replications is necessary.

Thus, the methodological difference is necessary because:

- SLLN checks **path wise convergence**
- CLT checks **distributional convergence**

4. Methodology

4.1 Programming Environment

All simulations were implemented using **Python**. The main libraries used in this project include:

- **NumPy**: random number generation and numerical operations,
- **Matplotlib**: plotting convergence graphs and histograms,
- **SciPy (scipy.stats)**: probability plots (Normal Q-Q plots) and normal density functions.

4.2 Random Number Generation

Random samples were generated using pseudo-random number generators provided by NumPy.

To ensure reproducibility of results, a fixed random seed was used when running the final simulations. Each simulation relies on iid. sampling, which is consistent with the assumptions required for both SLLN and CLT.

4.3 Simulation Parameters

SLLN Simulation Parameters

- Distribution: $X_i \sim U(0,1)$
- Theoretical mean: $\mu = 0.5$
- Theoretical variance: $\sigma^2 = \frac{1}{12}$
- Number of observations: $n = 10000$

The cumulative sample mean was computed as:

$$S_n = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

for all $n = 1, 2, \dots, 10000$, and the convergence of S_n to μ was visualized using a line plot.

CLT Simulation Parameters

For the CLT experiment, multiple sample sizes were considered:

$$n \in \{2, 5, 10, 30, 50\}$$

For each n , the number of replications was:

$$m = 1000$$

In each replication i , we generated n samples from $U(0, 1)$, computed the sum $S_n^{(i)}$, and then standardized it:

$$Z_i = \frac{S_n^{(i)} - n\mu}{\sigma\sqrt{n}}$$

The distribution of the standardized values was visualized using:

histograms with overlaid $N(0, 1)$ density curve,

Normal Q-Q plots.

Monte Carlo π Estimation Parameters

For Monte Carlo π estimation:

- Random points (x, y) were sampled uniformly from the unit square $[0,1] \times [0,1]$
- The point was classified as inside the quarter unit circle if:

$$x^2 + y^2 \leq 1$$

- The estimate after k points was computed as:

$$\hat{\pi}_k = 4 \cdot \frac{N_k}{k}$$

where N_k is the cumulative number of points satisfying the circle condition.

The convergence of $\hat{\pi}_k$ toward the true value of π was plotted for $k = 1, \dots, n$.

5. Results

5.1 SLLN Results

The SLLN convergence plot (Figure 1) demonstrates that the cumulative sample mean starts with high variability for small sample sizes. As n increases, the fluctuations decrease and the curve stabilizes around the theoretical mean $\mu = 0.5$.

A key observation is that the convergence becomes visually stable after a sufficiently large n . In particular, after several thousand observations, the cumulative mean remains close to 0.5 and large deviations become rare. This behavior provides empirical confirmation of the **almost sure convergence** predicted by the SLLN.

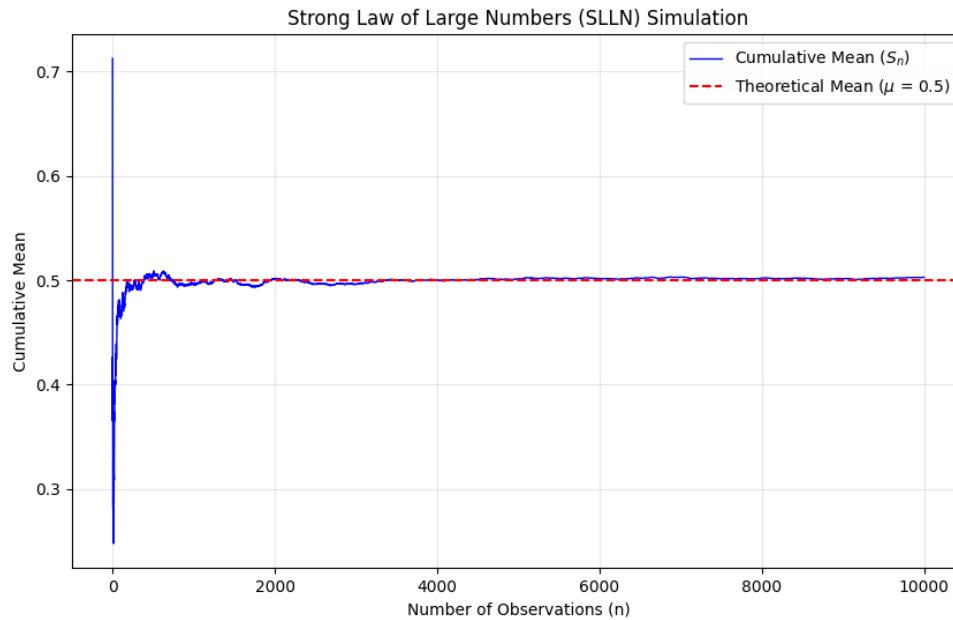


Figure 1: SLLN simulation – convergence of the cumulative sample mean to $\mu = 0.5$.

Convergence rate observation:

While the SLLN guarantees convergence, it does not provide an explicit universal rate. Experimentally, the variance in the sample mean decreases with increasing n , consistent with the idea that:

$$\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$$

Thus, larger sample sizes yield more stable averages.

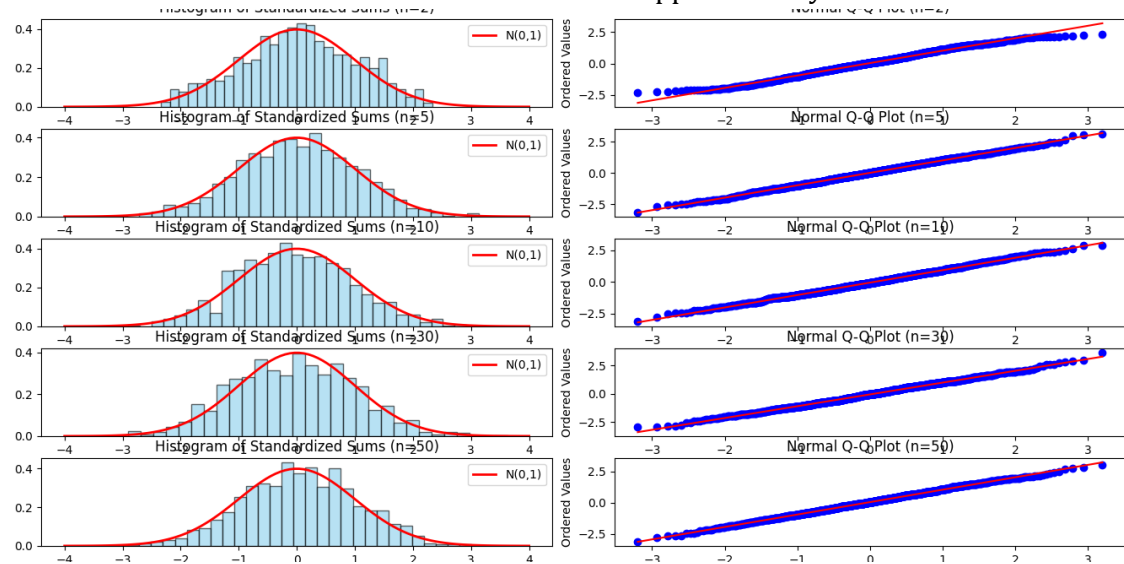
5.2 CLT Results

The CLT results were analyzed for sample sizes $n = 2, 5, 10, 30, 50$ using $m = 1000$ replications.

For small sample sizes (e.g., $n = 2$ and $n = 5$), the histograms show clear deviations from the standard normal curve. The distributions are less symmetric and show heavier irregularities, indicating that the normal approximation is not accurate at very low n .

As n increases (especially at $n = 30$ and $n = 50$):

- the histogram aligns closely with the theoretical $N(0,1)$ density curve,
- Q-Q plots show points increasingly falling along the reference line, which indicates that the standardized sums become approximately normal.



Figures 2–6: CLT simulation histograms and Q-Q plots for varying n .

Rate of convergence:

The convergence to normality becomes more visually apparent as n increases. In this simulation, $n \geq 30$ appears to provide a reasonably good approximation to normality for the Uniform(0,1) distribution.

5.3 Monte Carlo π Estimation Results

The Monte Carlo estimation plot (Figure 7) shows the estimated value $\hat{\pi}_k$ converging toward the true value of π as the number of simulated points increases.

At the beginning (small k), the estimate fluctuates significantly, because early sampling has high variance. As k grows, fluctuations decrease and the estimate stabilizes around π . This is

consistent with the SLLN, since the estimator is based on a sample proportion.

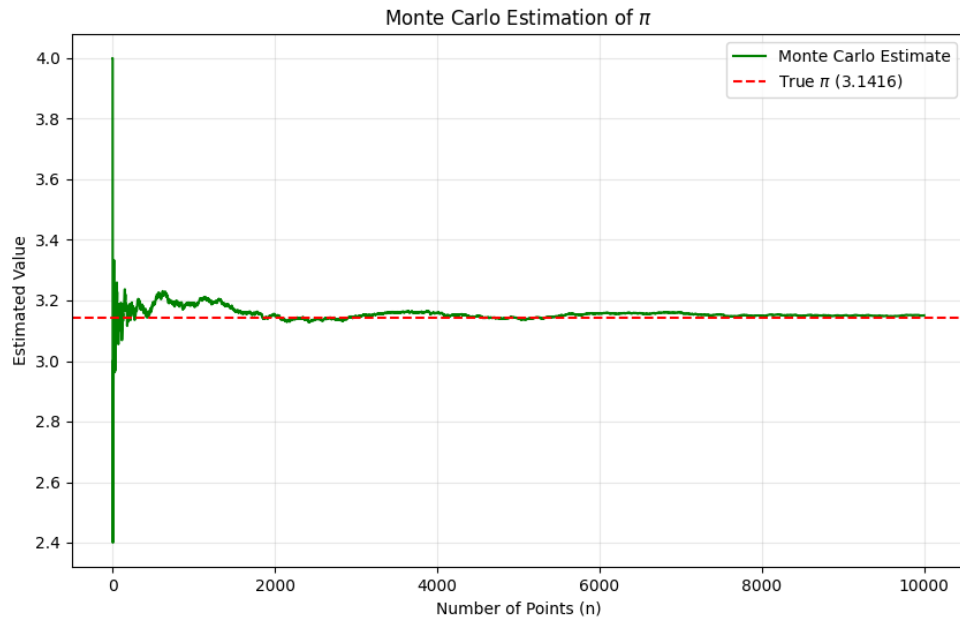


Figure 7: Monte Carlo estimation of π – convergence of $\hat{\pi}_k$ to π .

Error analysis:

The estimation error can be expressed as:

$$| \hat{\pi}_k - \pi |$$

The plot indicates that the error decreases with k , although it does not decrease monotonically. This is expected, since Monte Carlo estimates have stochastic variability. Overall, the long-run behavior supports convergence to the true value.

6. Discussion and Conclusion

This project provided empirical verification of two major theorems in probability theory: the Strong Law of Large Numbers and the Central Limit Theorem. The simulations clearly demonstrate how increasing sample size affects both the stability of sample averages and the shape of sampling distributions.

6.1 How large should n be?

There is no single universal value of n that guarantees “good enough” convergence in every setting. However, based on our simulations:

- For **SLLN**, convergence of the sample mean becomes visually stable after several thousand observations. In our experiment with $n = 10000$, the cumulative mean remained close to 0.5 in later stages.
- For **CLT**, normality becomes a reasonable approximation when n is moderately large. In this project, the results suggest that $n \geq 30$ produces distributions that closely resemble $N(0,1)$.

Thus, a practical answer is:

- **SLLN**: n should be in the thousands to obtain a stable mean.
- **CLT**: n around 30 or above often provides acceptable normal approximation.

6.2 Comparison of SLLN and CLT Convergence Rates

The SLLN convergence is observed along a single sample path, and stability increases gradually as more observations are collected. In contrast, the CLT convergence concerns the distributional behavior of standardized sums, requiring repeated experiments to observe the limiting distribution.

Experimentally, CLT convergence to normality can appear “faster” in the sense that moderate n values (e.g., 30) already yield approximately normal distributions. However, this does not contradict the nature of SLLN, since they address different mathematical objects and convergence concepts.

6.3 Practical Implications of Different Modes of Convergence

A major takeaway from this project is the methodological implication of convergence type:

- **Almost sure convergence (SLLN)** justifies using large samples to estimate expected values reliably in engineering and business contexts.
- **Convergence in distribution (CLT)** enables normal approximations in inference, confidence intervals, quality control, and statistical modeling.

Additionally, Monte Carlo π estimation illustrates how these theoretical foundations can be used for numerical approximation in practice, reinforcing the importance of simulation-based methods.

6.4 Final Conclusion

Overall, the simulation outcomes align strongly with theoretical expectations:

- the sample mean converges to μ as predicted by SLLN,
- standardized sums approach normality as predicted by CLT,
- Monte Carlo estimation converges to π as an application of law of large numbers.

This project highlights the power of simulation to both validate probability theory and develop intuition about randomness, convergence, and approximation.